

OVERVIEW

Donor Management System

A Laravel-based platform for managing donors, students, projects, and donations.

Laravel · MySQL · Queue Workers · Spatie Activitylog · Meta WhatsApp Cloud API



Donors



Students



Projects



Donations



Project Overview

The system centralizes the full charitable workflow — from registering donors and beneficiary students to managing projects and recording every donation.



Donors

Contact profiles, sources & giving preferences



Students

Beneficiary records with funding status



Projects

Campaigns with budgets & progress tracking



Donations

Receipts, payment methods, Gift Aid & status

Donor Mgmt

Dashboard

Projects

Project Types

Donors

Donor Sources

Students

Donations

Expenses

Notifications

Reports

Users & Roles

Recycle Bin

Activity Logs

Dashboard

Welcome back, Admin User!

Here's a snapshot of the system.

Total Donations

\$ 800 BDT

Total Expenses

₹ 398,500 BDT

Cash on Hand

₹ -397,700 BDT

Budget Allocated

₹ 3,200,000 BDT

Projects

5

Donors

2

Students

3

Unmatched Txns

0

Top Projects (by funded %)

Project	Status	Raised	Spent	Funded
Ramadan Food Distribution 2026	active	700 BDT	110,000 BDT	0.09%
Madrasah Scholarship Program	active	100 BDT	68,500 BDT	0.01%
Emergency Medical Aid Fund	active	0 BDT	18,000 BDT	0%
Tube-well Installation - Rural	pending	0 BDT	0 BDT	0%
Winter Blanket Drive 2025	completed	0 BDT	202,000 BDT	0%

Top Donors

#	Donor	Contributed
1	Arif Rahman DNR-1001	400 BDT
2	Rahim Ahmed DNR-1002	400 BDT

Admin User
admin

Live system dashboard — financial KPIs, top projects and top donors at a glance



Database Architecture

Relational & Normalized Schema

A MySQL schema built around clearly defined entity relationships, with referential integrity enforced at the database layer.



Foreign-Key Integrity

- Migrations carefully ordered to resolve table dependency chains
- Cascade rules and soft-delete columns applied across relations

CORE TABLES



users



roles



students



projects



project_types



donors



donor_sources



donations



expenses



activity_log



whatsapp_templates



notifications

12 interrelated tables · normalized to reduce redundancy



Role-Based Access Control (RBAC)

Every route and action is gated by role — access scopes are enforced through middleware and policy checks.

super_admin

Full system control & configuration access

admin

Manages all modules and operational data

accountant

Donations, expenses & financial reporting

user

Field volunteer — limited data-entry rights

HOW ACCESS IS ENFORCED



Authenticated request

User must be signed in



Role middleware

Checks the user's assigned role



Policy gate

Authorizes the specific action



Access granted

Otherwise the request is denied



Projects Module

Full lifecycle management of charitable projects and campaigns.

Donor Mgmt

Dashboard

Projects

Project Types

Donors

Donor Sources

Students

Donations

Expenses

Notifications

Reports

Users & Roles

Recycle Bin

Activity Logs

Projects

Admin User
admin

Projects

+ Add Project

Q Search by name or code

Status

Project Type

Reset

ID	Project Code	Name	Type	Budget	Status	Actions
1	PRJ-RAMADAN-2026	Ramadan Food Distribution 2026	Zakat	750,000 BDT	active	Edit Delete
2	PRJ-EDU-SCHOLAR	Madrasah Scholarship Program	Education	1,200,000 BDT	active	Edit Delete
3	PRJ-MED-EMRG	Emergency Medical Aid Fund	Medical	500,000 BDT	active	Edit Delete
4	PRJ-WINTER-2025	Winter Blanket Drive 2025	Sadaqah	300,000 BDT	completed	Edit Delete
5	PRJ-WATER-001	Tube-well Installation - Rural	Others	450,000 BDT	pending	Edit Delete

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- Validated create, edit and delete operations
- Status workflow: pending, active, completed
- Linked to donations, expenses and reports



Project Types Module

Reusable categories used to classify every project.

Donor Mgmt

Dashboard

Projects

Project Types

Donors

Donor Sources

Students

Donations

Expenses

Notifications

Reports

Users & Roles

Recycle Bin

Activity Logs

Project Types

Admin User

admin

Project Types

+ Add Type

Q Search by name

Status

Reset

ID	Name	Status	Actions
1	Zakat	active	Edit Delete
2	Sadaqah	active	Edit Delete
3	Education	active	Edit Delete
4	Medical	active	Edit Delete
5	Others	active	Edit Delete

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Donors Module

Complete donor relationship records with notification preferences.

Donor Mgmt

Dashboard

Projects

Project Types

Donors

Donor Sources

Students

Donations

Expenses

Notifications

Reports

Users & Roles

Recycle Bin

Activity Logs

Donors

Admin User
admin

Donors

+ Add Donor

Q Search by name, donor ID, phone, or post code

Preferred Project

Reset

Donor ID	Name	Contact	Address Line 1	Post Code	Source	Preferred Project	Donations	Notify
DNR-1001	Arif Rahman		22 Whitechapel Road, London	E1 1EW	Website	Ramadan Food Distribution 2026	2 times 300 BDT	Email SMS WhatsApp
DNR-1002	Rahim Ahmed		House 12, Road 5, Dhaka	1209	Facebook Campaign	—	0 times 0 BDT	Email SMS WhatsApp

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Create New Donor

Donor ID will be auto-generated (DNR-1001, DNR-1002, ...)

* Name

Donor full name

* Phone

+4471XXXXXXX

Email

optional@example.com

* Post Code

E1 1EW

* Address Line 1

Address Line 2

* City

Country

United Kingdom

Source

Select source

Preferred Project

Optional

Address Line 3

Notification channels

☒ Email

☒ SMS

☒ WhatsApp

Cancel

Save Change



Donor Sources Module

Tracks the acquisition channels through which donors are reached.

The screenshot shows the 'Donor Sources' module interface. On the left, there's a list of sources: Website, Facebook Campaign, Bank Transfer, and Event/Fundraiser. A modal titled 'Create Donor Source' is open, showing fields for Name (e.g., Website, Facebook Campaign), Description, and an Active toggle (No/Yes). On the right, a table displays the sources with columns for Donors, Enabled, and Actions (Edit, Delete).

Name	Donors	Enabled	Actions
Website	1	Yes	Edit Delete
Facebook Campaign	1	Yes	Edit Delete
Bank Transfer	0	Yes	Edit Delete
Event/Fundraiser	0	No	Edit Delete

LIST Sources — Website, Facebook, Bank Transfer, with donor counts

CREATE New source — name, description and active flag

The screenshot shows the 'Edit Source: Website' modal. It contains fields for Name (Website), Description (Donors who registered directly through our main web portal.), and an Active toggle (No/Yes). Buttons for Cancel and Update Source are at the bottom.

EDIT Update channel details and availability

Capabilities

- Records how each donor was acquired
- Counts donors attributed to each channel
- Supports campaign performance analysis



Students Module

Beneficiary student records linked to funding and donations.

Student ID	Name	Guardian	Education	Post Code	Funding	Donations	Actions
STD-2001	Aisha Rahman	Fazlur Rahman +447123456789	Secondary Oaklands School	E1 1DU	fully funded	1	View Edit Delete
STD-2002	Zayan Malik	Yousuf Malik +447987654321	Primary Nelson Mandela School	B1 1AY	partially funded	0	View Edit Delete
STD-2003	Maryum Khan	Tariq Khan +447555666777	Higher Manchester Academy	M1 1AE	unfunded	0	View Edit Delete

LIST Students with guardian, education and funding status

Create New Student

Student ID will be auto-generated (STD-2001, STD-2002, ...)

* Student Name

Guardian Name

Guardian Phone

Address

Post Code

Educational Level

Institution

Funding Status

Cancel Save Change

CREATE New student — auto ID, guardian and education details

Edit Student: STD-2001

* Student Name

Guardian Name

Guardian Phone

Address

Post Code

Educational Level

Institution

Funding Status

Cancel Update Student

EDIT Update beneficiary record and funding status

Aisha Rahman (STD-2001)

Student ID	STD-2001
Name	Aisha Rahman
Guardian	Fazlur Rahman - +447123456789
Address	123 Whitechapel Road, London
Post Code	E1 1DU
Education	Secondary - Oaklands School
Funding	fully funded
Donations Received	1

VIEW Student profile with donations received to date



Donations Module

Recording and management of every donation transaction.

The screenshot displays the 'Donations Module' interface. On the left is a dark sidebar with a 'Donor Mgmt' section containing links to Dashboard, Projects, Project Types, Donors, Donor Sources, Students, Donations (highlighted), Expenses, Notifications, Reports, Users & Roles, Recycle Bin, and Activity Logs. The main area shows a 'Record New Donation' modal form. The form includes a notification: 'Receipt number will be auto-generated (REC-100001, REC-100002, ...)'. It has fields for Donor, Project, Link to Student (optional), Amount (\$ 0.00), Payment Method (Card Payment), Transaction Date (2026-05-23 14:15:41), and Status (Confirmed). There are also checkboxes for Gift Aid and Consent, both currently set to 'No'. A note states: 'I am a UK taxpayer and I wish (Inspired By Islam) to reclaim tax back on all donations I have made within the last 6 years and all donations that I make hereafter. *'. At the bottom are 'Cancel' and 'Save Donation' buttons. To the right of the modal, a table shows a list of donations with columns for Consent, Status, and Actions. The table has two rows, both with 'No' for Consent and 'confirmed' for Status. The Actions column contains 'Edit' and 'Delete' icons. A pagination bar at the bottom shows '< 1 >'. The top right corner shows the user 'Admin User' with the role 'admin'.

Record New Donation

Receipt number will be auto-generated (REC-100001, REC-100002, ...)

* Donor: Select donor

* Project: Select project

Link to Student (optional): Select student

* Amount: \$ 0.00

* Payment Method: Card Payment

* Transaction Date: 2026-05-23 14:15:41

Status: Confirmed

Gift Aid: No

I am a UK taxpayer and I wish (Inspired By Islam) to reclaim tax back on all donations I have made within the last 6 years and all donations that I make hereafter. *

Consent: No

Donor give consent for personal data to be used for future communication, but not to share with third party

Cancel Save Donation

Select Campaign: Select Campaign

On-Off Recurring

Other: 0

Zakat Fitra Sadaqah

0 BDT

Consent	Status	Actions
No	confirmed	Edit Delete
No	confirmed	Edit Delete

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Capabilities

- Auto-generated unique receipt numbers
- Supports one-off and recurring donations
- Captures Gift Aid and donor consent



Expenses Module

Project-linked expenditure tracking with approval workflow.

The screenshot shows the 'Expenses' module interface. On the left is a sidebar with navigation links: Dashboard, Projects, Project Types, Donors, Donor Sources, Students, Donations, Expenses (selected), Notifications, Reports, Users & Roles, Recycle Bin, and Activity Logs. The main area displays the 'Expenses' section with a summary: 'Total Spent (filtered)' at \$398,500 BDT and 'Expense Count' at 9. Below this are filters for Project, Category, Status, Start date, and End date, along with a search bar for 'Vendor / description'. A table lists expenses with columns: Date, Project, Category, Amount, Vendor, Status, Recorded by, and Actions. The table contains 8 rows of data, including entries for Emergency Medical Aid Fund, Madrasah Scholarship Program, and Ramadan Food Distribution.

Date	Project	Category	Amount	Vendor	Status	Recorded by	Actions
2026-05-10	Emergency Medical Aid Fund - PRJ-MED-EMRG	Equipment	18,000 BDT	MedSupply BD	paid	Admin User	Edit Delete
2026-05-03	Madrasah Scholarship Program - PRJ-EDU-SCHOLAR	Supplies	8,500 BDT	Boi Bazar	paid	Admin User	Edit Delete
2026-04-30	Madrasah Scholarship Program - PRJ-EDU-SCHOLAR	Salaries	60,000 BDT	—	paid	Admin User	Edit Delete
2026-04-15	Ramadan Food Distribution 2026 - PRJ-RAMADAN-2026	Salaries	25,000 BDT	—	approved	Admin User	Edit Delete
2026-04-08	Ramadan Food Distribution 2026 - PRJ-RAMADAN-2026	Logistics	12,000 BDT	Dhaka Pickup Service	paid	Admin User	Edit Delete
2026-04-05	Ramadan Food Distribution 2026 - PRJ-RAMADAN-2026	Supplies	28,000 BDT	Halal Foods Ltd	paid	Admin User	Edit Delete
2026-04-02	Ramadan Food Distribution 2026 - PRJ-RAMADAN-2026	Supplies	45,000 BDT	Karim Wholesale Rice	paid	Admin User	Edit Delete

LIST Expenses with project, category, vendor and status

The screenshot shows the 'Record New Expense' modal form. It includes fields for: Project (dropdown), Category (dropdown with example 'e.g. Supplies'), Amount (BDT) (input field with 0.00), Expense Date (calendar icon), Status (dropdown with 'Approved' selected), Vendor / Payee (optional) (input field with example 'e.g. Karim Wholesale Rice'), and Description (optional) (text area with placeholder 'What was this expense for?'). There are 'Cancel' and 'Save Expense' buttons at the bottom.

CREATE Record expense — project, category, amount and vendor

The screenshot shows the 'Edit Expense' modal form. It includes fields for: Project (dropdown with 'PRJ-MED-EMRG - Emergency Medical Aid Fund'), Category (dropdown with 'Equipment'), Amount (BDT) (input field with 18000), Expense Date (calendar icon), Status (dropdown with 'Paid' selected), Vendor / Payee (optional) (input field with 'MedSupply BD'), and Description (optional) (text area with 'Emergency first-aid kits (50 units)'). There are 'Cancel' and 'Update Expense' buttons at the bottom.

EDIT Update spending records and approval status

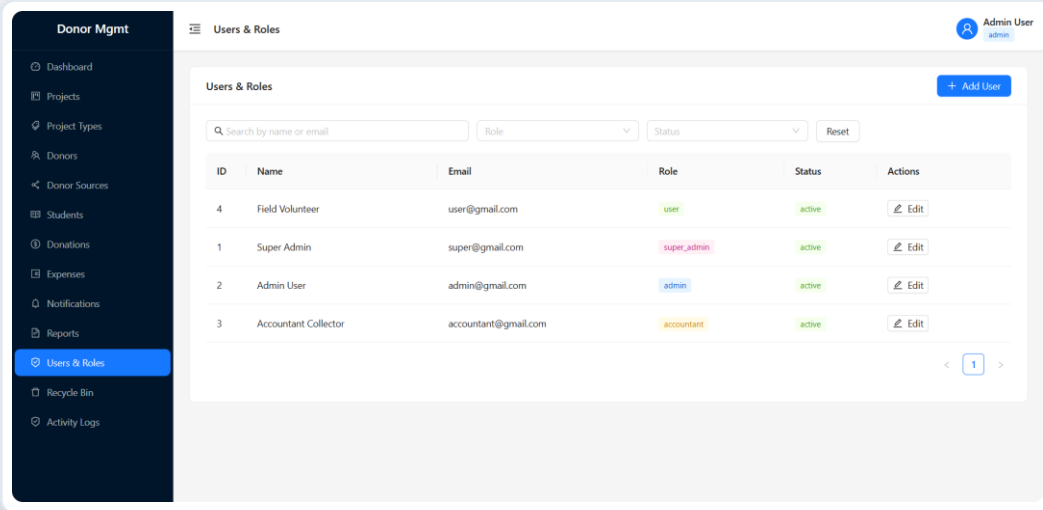
Capabilities

- Every expense linked to a specific project
- Approval status workflow for accountability
- Feeds directly into the financial reports

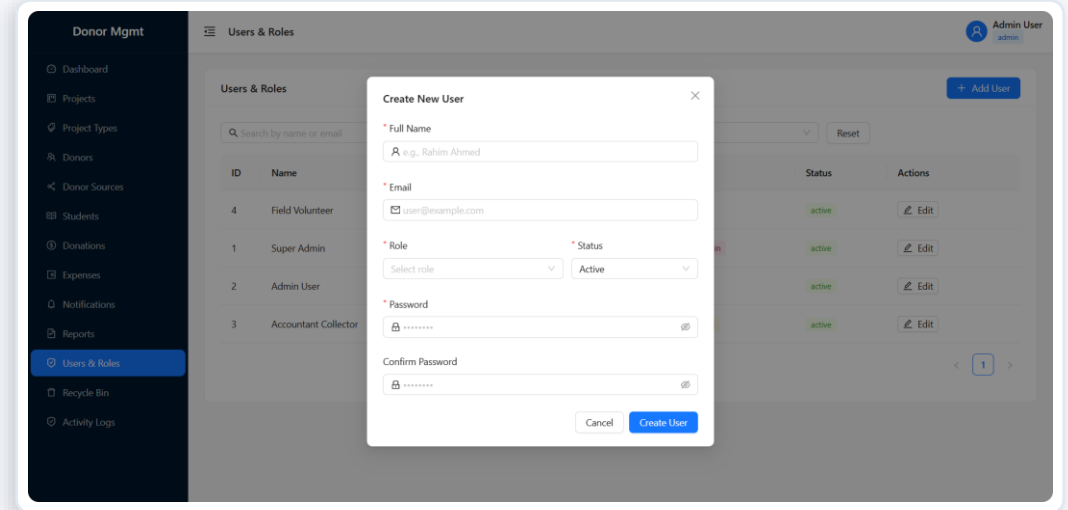


Users & Roles Module

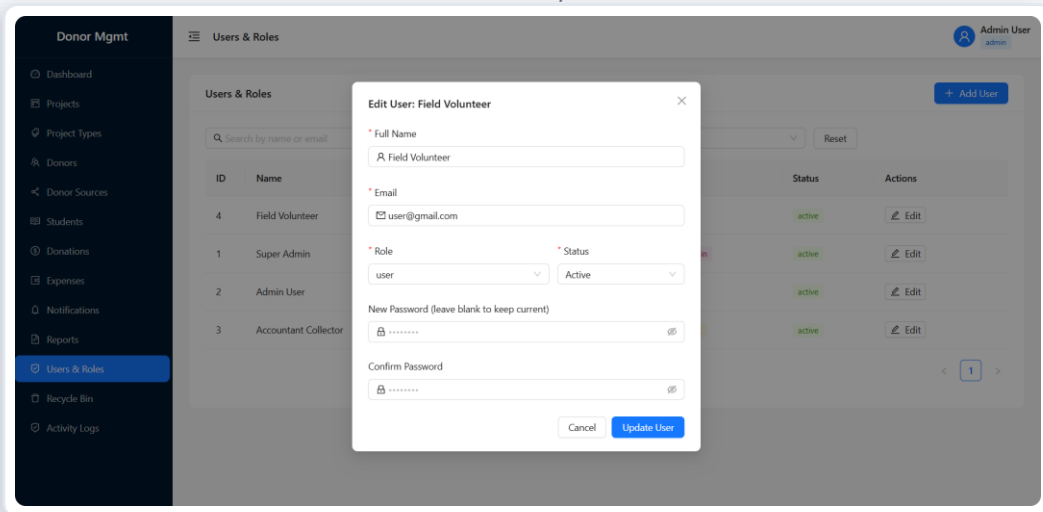
Administration of system accounts and their access levels.



LIST User accounts filtered by role and account status



CREATE Create user — assign role, status and credentials



EDIT Change a user's role or reset their password

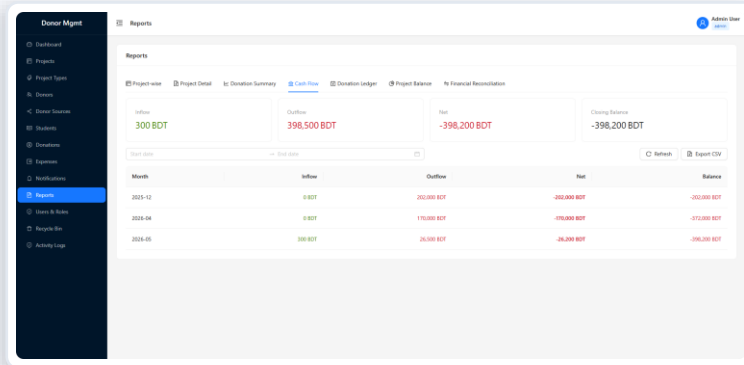
Capabilities

- Four-tier role system for least-privilege access
- Account activation and deactivation control
- Directly drives RBAC enforcement



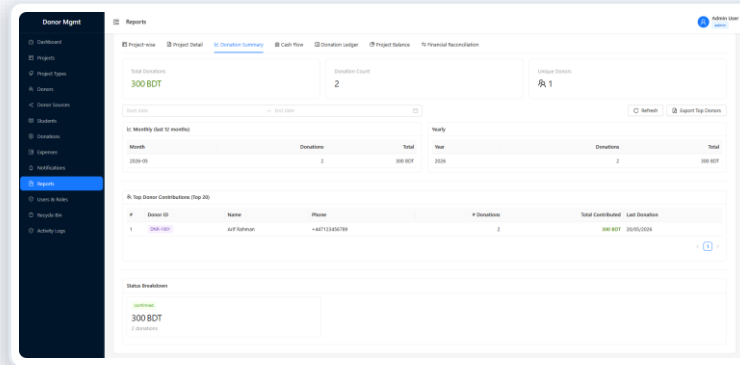
Reports & Financial Analytics

A reporting suite turning donation and expense data into actionable financial insight.



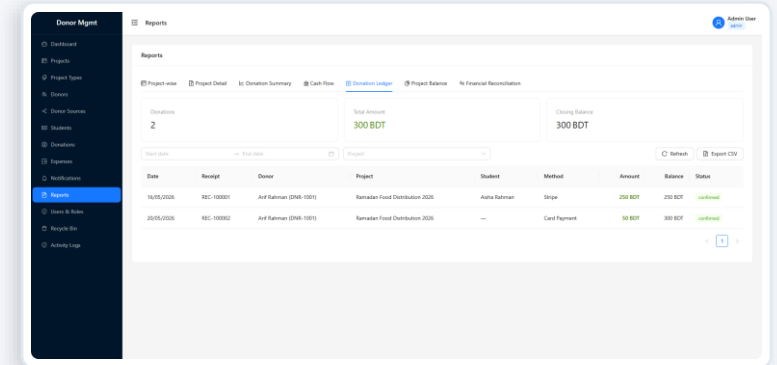
CASH FLOW

Inflow, outflow, net and closing balance by month



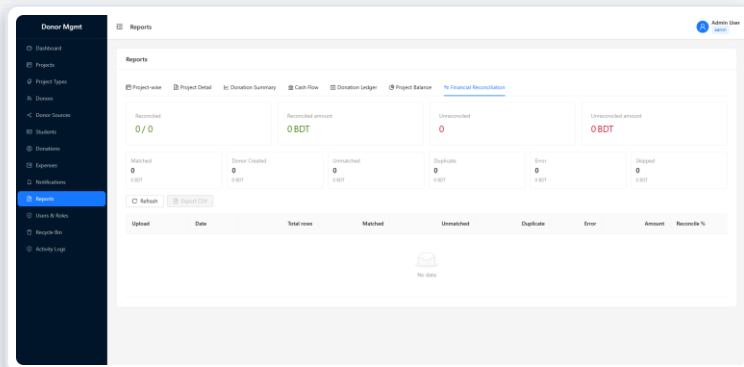
DONATION SUMMARY

Totals, unique donors and top contributors



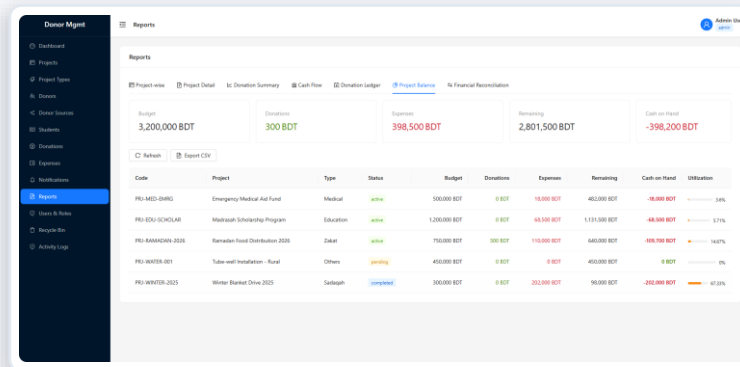
DONATION LEDGER

Itemized record of every donation made



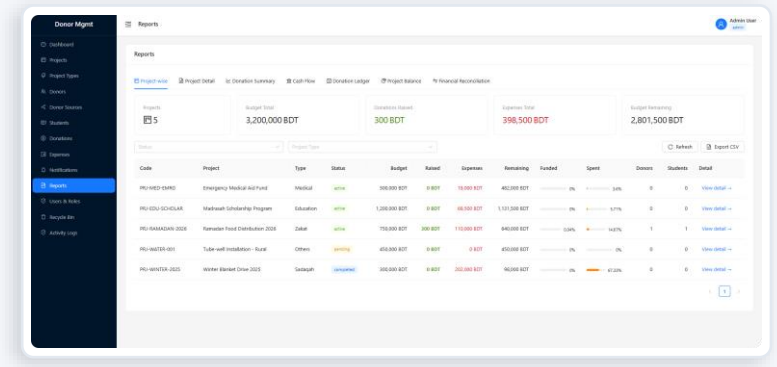
RECONCILIATION

Matched vs unmatched & duplicate transactions



PROJECT BALANCE

Budget, spend, remaining and utilization per project



PROJECT-WISE

Funding and beneficiary breakdown by project



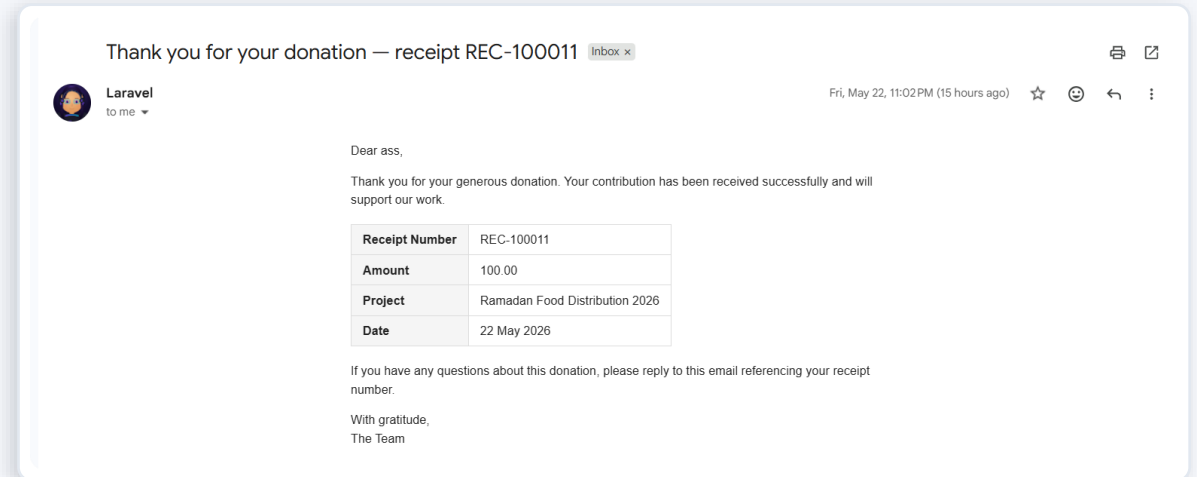
Scheduled Tasks & Email Notifications

Time-based and background work is automated through Laravel's task scheduler and queue workers.

Automated Background Work

- Donation receipt emails dispatched automatically
- Recurring donations processed on a fixed schedule
- Heavy operations offloaded to queued background jobs

```
$ php artisan queue:work
```



Donation receipt email — generated and queued by a scheduled job

PROCESSING PIPELINE



Scheduler

Triggers task on schedule



Job Queued

Work pushed to the queue



Worker Runs

Email sent / donation processed



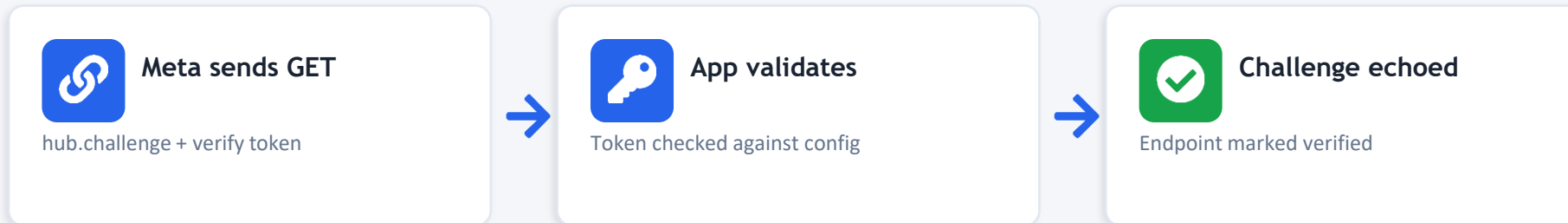
WhatsApp Notification Integration

Built on the Meta WhatsApp Cloud API

- Webhook endpoint verified via Meta's hub.challenge handshake
- ngrok tunnels the local server so Meta can reach the webhook
- Template-based messages send donation confirmations to donors
- Delivery status and inbound replies received through the webhook

```
$ ngrok http 8000
```

WEBHOOK VERIFICATION HANDSHAKE



Live donor notifications



Soft Deletes & Recycle Bin

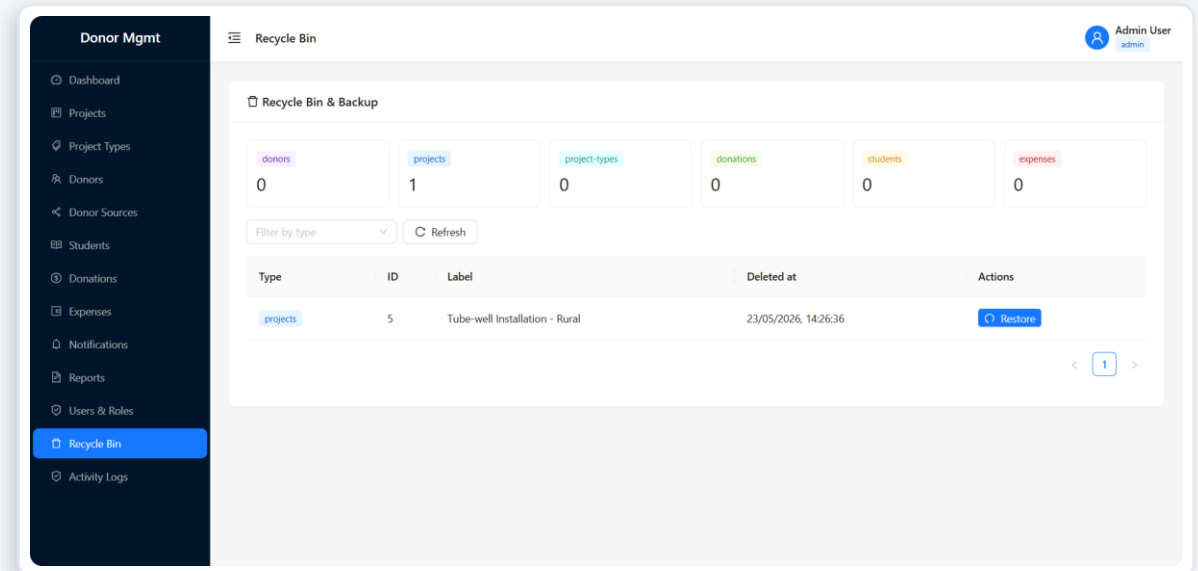
Records are never lost — Recoverable Deletion Across Every Module

- Deleting a record flags it rather than erasing it from the database
- The Recycle Bin groups removed records by type with live counts
- Authorized users can restore any item with a single click
- Protects against accidental data loss and supports recovery



Laravel SoftDeletes

Implemented with Eloquent's SoftDeletes trait — a `deleted_at` timestamp keeps records queryable and restorable.



Recycle Bin — soft-deleted records grouped by type, each with one-click restore



Event-Driven Architecture & Backend Logic



Events & Listeners

Domain events fire listeners that run side-effects in the background.

EVENT DonorRegistered



LISTENER Sends a welcome WhatsApp alert to the donor

EVENT DonationFailed



LISTENER Notifies an administrator of the failure



Optimized Backend APIs

Controllers expose clean, performant JSON endpoints to the front end.

- Activity-log controller with filtering and pagination
- Relations eager-loaded to avoid N+1 query overhead
- Consistent, lightweight JSON response structure

```
$ GET /api/activity-logs?event=created

{
  "data": [ { "event": "created", ... } ],
  "meta": { "page": 1, "total": 248 }
}
```



Project Status & Next Steps

12

Database Tables

9

Functional Modules

4

User Roles

3

Live Integrations

DELIVERED

- ✓ Role-based access control & user management
- ✓ Donor, student, project & donation modules
- ✓ Donor sources, project types & expenses
- ✓ Reporting suite with six financial views
- ✓ Soft deletes with Recycle Bin restore
- ✓ Spatie-powered audit logging
- ✓ Scheduled tasks & queued background jobs
- ✓ WhatsApp integration & event-driven alerts

NEXT STEPS

- ➔ Add automated test coverage across modules
- ➔ Polish dashboard analytics & reporting views
- ➔ Harden payment-gateway & failure handling
- ➔ Production deployment with monitoring

Core build complete — enhancement & hardening work ongoing.